

Multiplicative Drift Analysis of MOEA/D with $T = 1$ on ONEMINMAX under One-Bit Prior Noise (Corrected Version)

1 Problem Setup

1.1 The OneMinMax Benchmark

The ONEMINMAX benchmark is a bi-objective optimization problem defined over $X = \{0, 1\}^n$ with objective function

$$g(x) = (f(x), n - f(x)),$$

where $f(x) = |x|_1$ is the number of ones. The Pareto front is $\{(k, n - k) \mid k \in [0..n]\}$.

1.2 One-Bit Prior Noise

Let $p \in [0, 1]$. The noisy evaluation $\tilde{x}$ of x is: with probability p , flip one uniformly random bit of x ; with probability $1 - p$, $\tilde{x} = x$. The noisy objective is $\tilde{g}(x) := g(\tilde{x})$, with $\tilde{f}(x) := f(\tilde{x})$. Each noisy evaluation is independent. We have

$$\tilde{f}(x) \in \{f(x) - 1, f(x), f(x) + 1\},$$

with $\Pr[\tilde{f}(x) = f(x)] = 1 - p$, $\Pr[\tilde{f}(x) = f(x) - 1] = p \cdot f(x)/n$, and $\Pr[\tilde{f}(x) = f(x) + 1] = p \cdot (n - f(x))/n$.

1.3 MOEA/D with $T = 1$: Tchebycheff Decomposition

We use $N + 1$ subproblems with weight vectors $\lambda^i = (i/N, 1 - i/N)$ for $i = 0, \dots, N$, and the Tchebycheff scalarization with **ideal point** $z^* = (0, 0)$:

$$g^i(x) = \max \left\{ \frac{i}{N} f(x), \frac{N - i}{N} (n - f(x)) \right\}.$$

This is minimized at $f(x) = k_i^*$ where k_i^* is the value of k minimizing $\max\{(i/N)k, ((N - i)/N)(n - k)\}$. Setting the two terms equal gives $k_i^* = \lfloor (N - i) \cdot n/N \rfloor$. In particular, subproblem 0 (with $\lambda^0 = (0, 1)$) targets $f = n$, and subproblem N (with $\lambda^N = (1, 0)$) targets $f = 0$.

Since $T = 1$, each subproblem maintains its own solution x^i independently. Mutation flips exactly one uniformly chosen bit.

Remark 1 (No reevaluation). *When a solution is accepted, its noisy objective value is stored. This stored value is reused in all future comparisons until the solution is replaced. We do not reevaluate existing solutions.*

Algorithm 1 MOEA/D with $T = 1$ on ONEMINMAX (without reevaluation)

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1: for  $i = 0$  to  $N$  do
2:    $x^i \leftarrow$  random element of  $\{0, 1\}^n$ 
3:    $w^i \leftarrow \tilde{g}^i(x^i)$  // store noisy scalar value
4: end for
5: for  $t = 0, 1, 2, \dots$  do
6:   for  $i = 0$  to  $N$  do
7:      $y \leftarrow$  flip one uniformly random bit of  $x^i$ 
8:      $v \leftarrow \tilde{g}^i(y)$  // noisy evaluation of offspring
9:     if  $v \leq w^i$  then
10:       $x^i \leftarrow y$ ;  $w^i \leftarrow v$ 
11:    end if
12:   end for
13: end for
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2 Analysis of the Extreme Subproblem

We analyze subproblem $i = N$ with $\lambda^N = (1, 0)$, which targets $f = 0$ (minimizing f). The Tchebycheff function reduces to

$$g^N(x) = f(x).$$

By symmetry, the identical argument applies to subproblem 0 targeting $f = n$.

2.1 State Variable: the Stored Value w_t

Let w_t denote the stored (noisy) scalar value of subproblem N at the start of generation t . The key structural property is:

Lemma 2 (Monotonicity). w_t is nonincreasing: $w_{t+1} \leq w_t$ for all t .

Proof. The offspring y replaces x^N only if its noisy value $v = \tilde{f}(y) \leq w_t$, in which case $w_{t+1} = v \leq w_t$. Otherwise $w_{t+1} = w_t$. $\square$

This makes w_t a natural potential for multiplicative drift. It is a function of the algorithm's state (not of an external event), it is Markov given the current solution, and it is monotone.

2.2 Relating w_t to the True Value

Let $k_t = f(x_t^N)$ be the true f -value. Since the stored value w_t was obtained by a single noisy evaluation of some solution with true value k_t (or a predecessor), we have:

$$w_t \in \{k_t - 1, k_t, k_t + 1\}. \tag{1}$$

More precisely, $w_t = \tilde{f}(x_t^N)$ computed at the time x_t^N was accepted. Since one-bit noise shifts f by at most 1, the stored value is within 1 of the current true value. Note also $k_t \geq 0$ and $w_t \geq -1$ in principle, but $w_t \geq 0$ since $\tilde{f}$ takes values in $[0..n]$.

An important consequence: $w_t = 0$ implies $k_t \in \{0, 1\}$. In particular, $k_t \leq w_t + 1$ always.

2.3 Drift of w_t

Lemma 3. *There exists a constant $\alpha > 0$ such that for $p \leq \alpha/n$ and $w_t = w \geq 1$,*

$$\mathbb{E}[w_t - w_{t+1} \mid w_t = w] \geq \frac{w}{4n}.$$

Proof. We need to lower-bound the expected decrease of the stored value. Since w_t is nonincreasing (Lemma 2), we have $w_t - w_{t+1} \geq 0$ always, so we only need to lower-bound the probability of a strict decrease and bound the magnitude.

Transition structure. The mutation flips exactly one bit, so the offspring y has true value

$$f(y) \in \{k_t - 1, k_t + 1\}.$$

The noisy evaluation then gives $\tilde{f}(y) \in \{f(y) - 1, f(y), f(y) + 1\}$, so

$$\tilde{f}(y) \in \{k_t - 2, k_t - 1, k_t, k_t + 1, k_t + 2\}.$$

The offspring is accepted iff $\tilde{f}(y) \leq w_t = w$.

Identifying the improving event. We aim for the event $w_{t+1} = w - 1$, i.e., $\tilde{f}(y) = w - 1 \leq w$ (so it is accepted). We construct a specific chain of events that achieves this.

Since $k_t \leq w + 1$ (from (1)) and $k_t \geq w - 1$, we consider cases.

Case A: $k_t = w$ (true value equals stored value).

Flip a 1-bit (prob $k_t/n = w/n$), giving $f(y) = w - 1$. Then y is correctly evaluated (prob $1 - p$), giving $\tilde{f}(y) = w - 1$. Since $w - 1 < w = w_t$, the offspring is accepted and $w_{t+1} = w - 1$. Probability:

$$\frac{w}{n}(1 - p).$$

Case B: $k_t = w - 1$ (true value is one less than stored value).

This means the solution was previously misevaluated upward. Now flip a 1-bit (prob $(w - 1)/n$), giving $f(y) = w - 2$. Then y is correctly evaluated (prob $1 - p$), giving $\tilde{f}(y) = w - 2 < w$, accepted. Here $w_{t+1} = w - 2$, which is even better (decrease of 2). Probability at least:

$$\frac{w - 1}{n}(1 - p).$$

Alternatively, flip a 0-bit (prob $(n - w + 1)/n$), giving $f(y) = w$. Correct evaluation gives $\tilde{f}(y) = w \leq w$, accepted with $w_{t+1} = w$ (no improvement). But if the noise flips a 1-bit of y (prob $p \cdot w/n$), we get $\tilde{f}(y) = w - 1$, accepted with decrease of 1.

Case C: $k_t = w + 1$ (true value exceeds stored value).

Flip a 1-bit (prob $(w + 1)/n$), giving $f(y) = w$. Correct evaluation gives $\tilde{f}(y) = w \leq w$, accepted with $w_{t+1} = w$ (no change). If instead noise shifts to $w - 1$: $\tilde{f}(y) = w - 1$, giving decrease of 1. If flip a 0-bit, $f(y) = w + 2$, and correct evaluation gives $\tilde{f}(y) = w + 2 > w$, rejected.

Lower bound. In Case A (which is the most common starting configuration and always has positive probability of occurring at initialization), the probability of decreasing w_t by at least 1 is at least $w(1 - p)/n$. Since $w_t - w_{t+1} \geq 0$ always and $w_t - w_{t+1} \leq w$ (the stored value cannot go below 0), we have

$$\mathbb{E}[w_t - w_{t+1} \mid w_t = w, k_t = w] \geq 1 \cdot \frac{w(1 - p)}{n} \geq \frac{w}{2n},$$

for $p \leq 1/2$.

For Case B, the improving probability is at least $(w-1)(1-p)/n$. For Case C, even though a direct w -decrease is harder, the probability that w stays the same or decreases is positive, and since $w_{t+1} \leq w_t$ always, the drift is nonnegative.

Unconditioning on k_t . We need to remove the conditioning on k_t . Since w_t is fully determined by the algorithm state and $k_t \in \{w-1, w, w+1\}$, we take the worst case. The worst case for the drift is Case C ($k_t = w+1$), where a direct decrease of w requires either:

- Flip a 1-bit to get $f(y) = w$, then noise flips a 1-bit of y : prob $\frac{w+1}{n} \cdot \frac{pw}{n}$. This is $O(pw^2/n^2)$, which is small.
- Flip a 1-bit to get $f(y) = w$, correctly evaluated to w , accepted with $w_{t+1} = w$ (no decrease, but no loss either).

In this worst case the drift is indeed nonnegative but could be 0 to leading order. However, Case C requires that the solution was previously misevaluated *downward* (stored value $w < k_t = w+1$). For this to persist, every subsequent accepted offspring must also be misevaluated. The probability that Case C persists for many consecutive steps without transitioning to Case A or B is exponentially small.

More precisely, consider any step where $k_t = w+1$. If we flip a 1-bit (prob $(w+1)/n \geq w/n$) to get $f(y) = w$, and y is correctly evaluated (prob $1-p$), then $\tilde{f}(y) = w \leq w$, so y is accepted. Now $k_{t+1} = w$ and $w_{t+1} = w$: we have transitioned to Case A. The probability of this escape is at least $w(1-p)/n$. In the next step, Case A applies with drift $w(1-p)/n$.

Therefore, for a two-step drift argument: starting from any $k_t \in \{w-1, w, w+1\}$, within two steps the expected total decrease of w satisfies

$$\mathbb{E}[w_t - w_{t+2} \mid w_t = w] \geq \frac{w(1-p)}{n} \cdot \frac{w(1-p)}{n} \cdot 1$$

in the worst case (Case C first escapes to Case A, then improves). However, this gives a multiplicative drift of order w^2/n^2 per two steps, which is not strong enough.

Cleaner approach: amortized single-step bound. Instead of case analysis on k_t , we give a direct lower bound. Regardless of $k_t \in \{w-1, w, w+1\}$:

- The number of 1-bits in x_t^N is $k_t \geq w-1$.
- Flip a 1-bit: prob $k_t/n \geq (w-1)/n$.
- The offspring has $f(y) = k_t - 1 \leq w$.
- Correct evaluation of offspring: prob $1-p$, giving $\tilde{f}(y) = k_t - 1 \leq w = w_t$, so accepted.
- If $k_t - 1 \leq w-1$, then $w_{t+1} = k_t - 1 \leq w-1$, a decrease of at least 1.
- If $k_t - 1 = w$, then $w_{t+1} = w$, no decrease.

The only problematic case is $k_t = w+1$ giving $f(y) = w$ and $w_{t+1} = w$. In the cases $k_t \in \{w-1, w\}$ (i.e., $f(y) \in \{w-2, w-1\}$), we get $w_{t+1} \leq w-1$.

So the probability of a strict decrease is at least:

$$\Pr[k_t \leq w] \cdot \frac{(w-1)(1-p)}{n}.$$

Bounding $\Pr[k_t \leq w]$. Since $w_t = \tilde{f}(x_t^N)$ was the noisy evaluation at acceptance time, $k_t = w + 1$ requires a downward noise event at that acceptance time: $\text{prob} \leq p$. More carefully, $k_t = w + 1$ iff the noise shifted k_t down by 1 to give stored value w , which required $p \cdot k_t/n \leq p$ at that moment. So

$$\Pr[k_t = w + 1 \mid w_t = w] \leq p.$$

Therefore $\Pr[k_t \leq w \mid w_t = w] \geq 1 - p$, and the probability of a strict decrease is at least

$$(1 - p) \cdot \frac{(w - 1)(1 - p)}{n} = \frac{(w - 1)(1 - p)^2}{n} \geq \frac{(w - 1)(1 - 2p)}{n}.$$

For $w \geq 2$ and $p \leq 1/(4n)$:

$$\mathbb{E}[w_t - w_{t+1} \mid w_t = w] \geq \frac{(w - 1)(1 - 2p)}{n} \geq \frac{w/2}{n} = \frac{w}{2n}.$$

For $w = 1$: we need $k_t = 1$ and flip the single 1-bit (prob $1/n$), correctly evaluated (prob $1 - p$), giving $\tilde{f}(y) = 0 < 1$, accepted. The probability is $(1 - p)/n \cdot (1 - p) \geq (1 - 2p)/n \geq 1/(2n)$. So

$$\mathbb{E}[w_t - w_{t+1} \mid w_t = 1] \geq \frac{1}{2n} \geq \frac{1}{4n}.$$

(We use $w/4n = 1/(4n)$ for $w = 1$.)

Combining all cases: for $p \leq \alpha/n$ with α sufficiently small,

$$\mathbb{E}[w_t - w_{t+1} \mid w_t = w] \geq \frac{w}{4n}. \quad \square$$

Remark 4. The bound $\Pr[k_t = w + 1 \mid w_t = w] \leq p$ deserves justification. At the moment x_t^N was accepted (say at time $s \leq t$), the stored value $w_s = \tilde{f}(x_s^N)$ was computed from a fresh noisy evaluation. Since no reevaluation occurs, $w_t = w_s$ and $k_t = k_s$ (the solution hasn't changed). The event $k_s = w_s + 1$ means $\tilde{f}(x_s^N) = f(x_s^N) - 1$, which under one-bit noise has probability $p \cdot f(x_s^N)/n \leq p$. This is a property of the acceptance-time noise, not a fresh draw, so it is well-defined and does not require reevaluation.

2.4 Hitting Time

Theorem 5. There exists $\alpha > 0$ such that for $p \leq \alpha/n$, the expected number of generations for subproblem N to reach $w_t = 0$ is

$$\mathbb{E}[T_N] = O(n \log n).$$

Proof. By Lemma 3, w_t satisfies multiplicative drift with $\delta = 1/(4n)$ and $w_0 \leq n$. The Multiplicative Drift Theorem [1] yields

$$\mathbb{E}[T_N] \leq \frac{1 + \ln(n)}{1/(4n)} = 4n(1 + \ln n) = O(n \log n). \quad \square$$

Remark 6. Reaching $w_t = 0$ guarantees $k_t \leq 1$, not necessarily $k_t = 0$. However, for the purpose of covering the Pareto front, having a solution with f -value in $\{0, 1\}$ at the extreme subproblem suffices to anchor the front. If one insists on $k_t = 0$: once $w_t = 0$, either $k_t = 0$ (done) or $k_t = 1$ (the stored value was a lucky downward noise). In the latter case, a single further step flipping the one 1-bit and evaluating correctly brings $k_t = 0$ and $\tilde{f}(y) = 0$, accepted. This costs $O(n)$ additional expected steps, absorbed into the $O(n \log n)$ bound.

By symmetry, $\mathbb{E}[T_0] = O(n \log n)$ for subproblem 0 targeting $f = n$.

3 Covering the Full Pareto Front

3.1 General Subproblems

For subproblem i with target k_i^* , the Tchebycheff function $g^i(x)$ decreases when $f(x)$ moves toward k_i^* from either side. Define the stored value $w_t^{(i)} = \tilde{g}^i(x_t^i)$. By the same acceptance rule, $w_t^{(i)}$ is nonincreasing. When $f(x_t^i)$ is far from k_i^* , moving one step closer strictly decreases g^i , and the noisy evaluation preserves the improvement with probability $(1 - p)^2$. The multiplicative drift argument of Lemma 3 adapts, yielding:

Lemma 7. *For each subproblem i and $p \leq \alpha/n$,*

$$\mathbb{E}[T_i] = O(n \log n) \quad \text{generations.}$$

3.2 Total Runtime

Theorem 8. *Let $N = n$ with weights chosen so that the g -optima cover the full Pareto front. For $p \leq \alpha/n$, the MOEA/D with $T = 1$ finds the complete Pareto front of ONEMINMAX under one-bit prior noise (without reevaluation) in expected*

$$\mathbb{E}[T_{\text{total}}] = O(n^2 \log n) \quad \text{function evaluations.}$$

Proof. We define one **generation** as one full sweep over all $N + 1$ subproblems, costing $N + 1$ function evaluations. Each subproblem converges in $O(n \log n)$ generations independently (Lemma 7). Therefore, the number of generations until all subproblems have converged is

$$\mathbb{E}\left[\max_i T_i\right] = O(n \log n).$$

Here we use the fact that the subproblems are independent (since $T = 1$), so the maximum of $n + 1$ independent random variables each with mean $O(n \log n)$ is $O(n \log n)$ in expectation. (More precisely, by a union bound and Markov's inequality: $\Pr[\max_i T_i > c \cdot n \log n] \leq (n + 1) \cdot \Pr[T_0 > c \cdot n \log n]$, which can be made arbitrarily small for large c using exponential tail bounds from the drift theorem.)

The total function evaluations are

$$(N + 1) \cdot O(n \log n) = (n + 1) \cdot O(n \log n) = O(n^2 \log n). \quad \square$$

4 Comparison with the SEMO Result

	SEMO (Dinot et al.)	MOEA/D, $T = 1$
Noise tolerance	$p \leq \alpha/n$	$p \leq \alpha/n$
Potential variable	$\ell_t = j_t - \mathbf{1}_{A_t} + 1$	w_t (stored scalar value)
Drift per iteration	$\Omega(k/(n \cdot \mathcal{P}_t))$	$\Omega(w/(n))$
Phase 1 (extremes)	$O(n^2 \log n)$ iterations	$O(n \log n)$ generations
Bottleneck	$1/ \mathcal{P}_t $ parent selection	None
Phase 2 (fill front)	$O(n^2)$ iterations	Not needed ($N = n$)
Total evaluations	$O(n^2 \log n)$	$O(n^2 \log n)$

The total evaluation count matches, but the sources of the n^2 factor differ:

- **SEMO:** Each of the $O(n^2 \log n)$ iterations involves one function evaluation. The quadratic factor arises because the probability of selecting the right parent from the population $\mathcal{P}_t$ is $1/|\mathcal{P}_t| = O(1/n)$.
- **MOEA/D, $T = 1$:** The algorithm runs for $O(n \log n)$ generations, each costing $n + 1$ evaluations. The quadratic factor comes from the parallelism overhead: $n + 1$ subproblems are updated per generation, even though each individual subproblem converges in $O(n \log n)$ of its own updates.

References

- [1] Benjamin Doerr, Daniel Johannsen, and Carola Winzen. Multiplicative drift analysis. *Algorithmica*, 64:673–697, 2012.
- [2] Matthieu Dinot, Benjamin Doerr, Ulysse Hennebelle, and Sebastian Will. Runtime analyses of multi-objective evolutionary algorithms in the presence of noise. In *IJCAI 2023*. ijcai.org, 2023.